

SKILL_WEEK-08

Task 1: To Detect variable references, expand values, handle undefined variables, update tokens, support nested variables, and test expansion logic.

```
UW PICO 5.09

    if (k == 0)
    {
        output[j++] = '$';
        continue;
    }

    char *value = getenv(var_name);

    if (value != NULL)
    {
        for (int m = 0; value[m] != '\0'; m++)
        {
            output[j++] = value[m];
        }
    }
    else
    {
        printf("Warning: Undefined variable %s\n", var_name);
    }
}
else
{
    output[j++] = input[i++];
}
}

output[j] = '\0';
}

int main()
{
    char input[MAX_INPUT];
    char expanded[MAX_OUTPUT];

    printf("Enter command: ");
    fgets(input, sizeof(input), stdin);

    input[strcspn(input, "\n")] = '\0';

    expand_variables(input, expanded);

    printf("\nOriginal Command : %s\n", input);
    printf("Expanded Command : %s\n", expanded);

    return 0;
}
```

Fig 1: C program for detecting and expanding environment variable references

Test Case - 1:

```
amrutha@Amruthas-MacBook-Air OSSP % gcc src/skill8_task1.c -o bin/skill8_task1
amrutha@Amruthas-MacBook-Air OSSP % ./bin/skill8_task1
Enter command: echo $USER

Original Command : echo $USER
Expanded Command : echo amrutha
amrutha@Amruthas-MacBook-Air OSSP %
```

Fig 2: Expansion of the USER environment variable in the command

Test Case - 2:

```
amrutha@Amruthas-MacBook-Air OSSP % ./bin/skill8_task1
Enter command: echo $HOME/Documents

Original Command : echo $HOME/Documents
Expanded Command : echo /Users/amrutha/Documents
amrutha@Amruthas-MacBook-Air OSSP %
```

Fig 3: Expansion of the HOME environment variable while preserving additional path text

Test Case - 3:

```
amrutha@Amruthas-MacBook-Air OSSP % ./bin/skill8_task1
Enter command: echo $USER $HOME

Original Command : echo $USER $HOME
Expanded Command : echo amrutha /Users/amrutha
amrutha@Amruthas-MacBook-Air OSSP %
```

Fig 4: Expansion of multiple environment variables in a single command

Test Case - 4:

```
amrutha@Amruthas-MacBook-Air OSSP % ./bin/skill8_task1
Enter command: echo $MYUNKNOWNVAR
Warning: Undefined variable $MYUNKNOWNVAR

Original Command : echo $MYUNKNOWNVAR
Expanded Command : echo
amrutha@Amruthas-MacBook-Air OSSP %
```

Fig 5: Handling an undefined environment variable during expansion

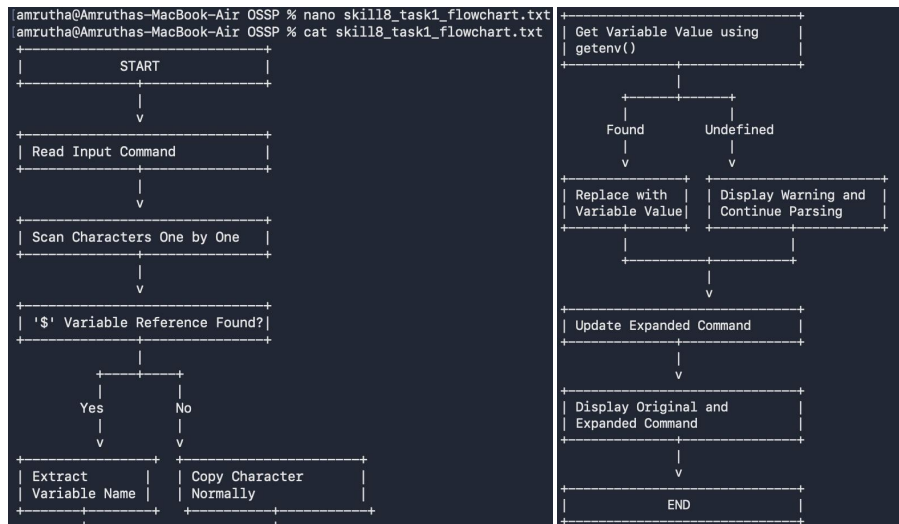


Fig 6: Flowchart for detecting and expanding environment variable references

Result:

Environment variable references were detected and expanded successfully. Variables such as \$USER and \$HOME were replaced with their corresponding values, multiple variables were handled correctly, and undefined variables were identified safely.

Task 2: Identify built-in commands, create a dispatch table, execute in-process commands, handle invalid commands, maintain state, and test dispatch logic.

```

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int main()
{
    char command[100];
    char argument[200];

    while (1)
    {
        printf("\nmini-shell> ");
        scanf("%99s", command);

        if (strcmp(command, "pwd") == 0)
        {
            builtin_pwd();
        }
        else if (strcmp(command, "cd") == 0)
        {
            scanf("%199s", argument);
            if (chdir(argument) == 0)
            {
                printf("Directory changed successfully.\n");
            }
            else
            {
                perror("cd failed");
            }
        }
        else if (strcmp(command, "help") == 0)
        {
            builtin_help();
        }
        else if (strcmp(command, "exit") == 0)
        {
            printf("Exiting mini shell...\n");
            break;
        }
        else
        {
            printf("Invalid built-in command: %s\n", command);
        }
    }

    return 0;
}

```

Fig 7: C program implementing built-in commands and command dispatch logic

Test Case - 1:

```
amrutha@Amruthas-MacBook-Air OSSP % gcc src/skill8_task2.c -o bin/skill8_task2
amrutha@Amruthas-MacBook-Air OSSP % ./bin/skill8_task2

mini-shell> pwd
Current Directory: /Users/amrutha/OSSP
```

Fig 8: Execution of the built-in pwd command to display the current working directory

Test Case - 2:

```
mini-shell> help
Available Built-in Commands:
pwd  - Display current directory
cd   - Change current directory
help - Display help information
exit - Exit the program
```

Fig 9: Display of supported built-in commands using the help command

Test Case - 3:

```
mini-shell> cd ..
Directory changed successfully.

mini-shell> pwd
Current Directory: /Users/amrutha
```

Fig 10: Changing and maintaining the current working directory using the built-in cd command

Test Case - 4:

```
mini-shell> hello
Invalid built-in command: hello
```

Fig 11: Handling an invalid built-in command in the mini shell

Test Case - 5:

```
mini-shell> exit
Exiting mini shell...
amrutha@Amruthas-MacBook-Air OSSP %
```

Fig 12: Termination of the mini shell using the built-in exit command

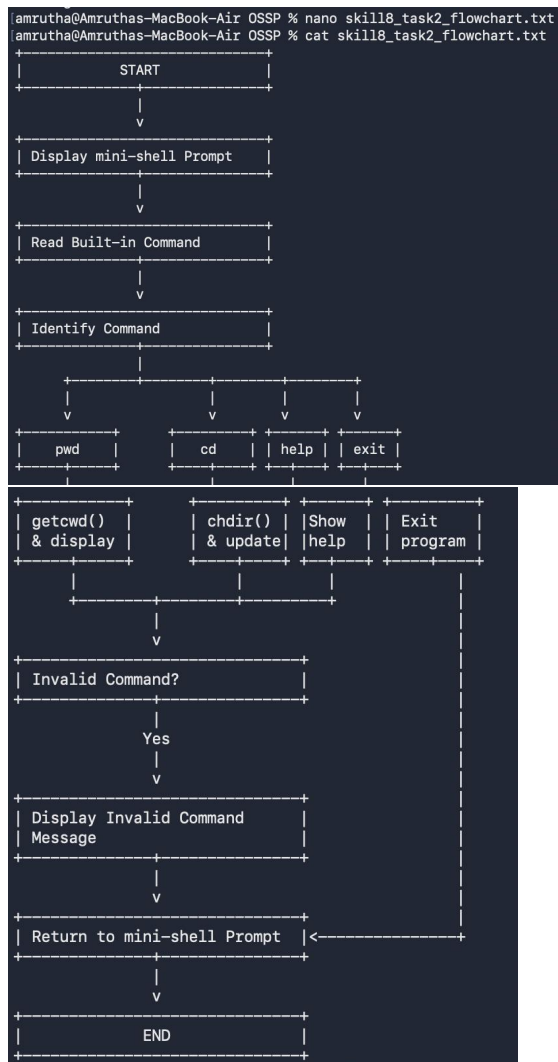


Fig 13: Flowchart for identifying and executing built-in commands in the mini shell

Result:

Built-in commands such as `pwd`, `cd`, `help`, and `exit` were implemented successfully. Commands were identified and executed within the same process, directory state was maintained using `cd`, and invalid commands were handled correctly.